Dynamics: Problem Sheet 5 (of 8)

1. A particle moves on the inside surface of a smooth cone with its axis vertical, defined by the equation $z = r$ in cylindrical polar coordinates (r, θ, z). Initially the particle is at height $z = a$, and its velocity is horizontal, speed v, in the $\mathbf{e}_\theta$ direction. Starting from Newton's second law show that $r^2\dot{\theta}$ is constant. Explain why the total energy is conserved, and deduce that

$$\dot{z}^2 + \frac{1}{2}\frac{a^2v^2}{z^2} + gz \;=\; \frac{1}{2}v^2 + ga \;.$$

   Hence show that the particle remains at all times between two heights, which should be determined.

2. A particle of mass m, moving under gravity, is disturbed from rest at the highest point on the outside of a smooth sphere of radius a.

   (a) Explain why the particle subsequently moves on a great circle.

   (b) By introducing plane polar coordinates in the vertical plane of this circle (or otherwise), show that

$$\ddot{\theta} \;=\; \frac{g}{a}\sin\theta \;, \qquad N \;=\; mg\cos\theta - ma\dot{\theta}^2 \;.$$

   Here $\theta(t)$ denotes the angle between the *upward* vertical axis of the sphere and the straight line from the particle to the centre of the sphere (the usual polar angle for a sphere), and N is the magnitude of the normal reaction.

   (c) Show that the normal reaction is given by

$$N \;=\; mg\left(\frac{3z}{a} - 2\right) \;,$$

   where z is the height of the particle above the centre of the sphere. At what height does the particle lose contact with the sphere?

3. A bead of mass m is free to slide on a smooth wire that is made to rotate at constant angular velocity ω about the vertical axis through a fixed point O on the wire. The wire is bent into the shape of a parabola, $z = r^2/2a$, where z is measured vertically upwards from O, and r is the horizontal distance from O.

   (a) Show that if $z(t)$ and $r(t)$ are the vertical and horizontal distances of the bead from O, then

$$m\left[(\ddot{r} - r\omega^2)\mathbf{e}_r + 2\omega\dot{r}\,\mathbf{e}_\theta + \ddot{z}\,\mathbf{e}_z\right] \;=\; -mg\,\mathbf{e}_z + \mathbf{N} \;,$$

   where $\mathbf{N}$ is the normal reaction.

   (b) Hence deduce that

$$(a^2 + r^2)\ddot{r} + r\dot{r}^2 \;=\; (a^2\omega^2 - ga)r \qquad\qquad (*) \;.$$

   (c) Show that $r = 0$ is an equilibrium point. The *linearized equation of motion* about $r = 0$ is

$$a^2\ddot{\xi} \;=\; (a^2\omega^2 - ga)\xi \;,$$

   where we have written $r = \xi$, and kept only the linear terms in $\xi, \dot{\xi}$ in equation $(*)$, in a Taylor expansion around $\xi = 0$. Discuss the stability of the equilibrium point.

4. A point particle moves on a circle of radius l in the (z, x) plane, centred on the origin.

(a) i. By introducing polar coordinates $(z, x) = (-r\cos\theta, r\sin\theta)$, show that the particle has acceleration

$$\ddot{\mathbf{r}} \;=\; -l\dot{\theta}^2\,\mathbf{e}_r + l\ddot{\theta}\,\mathbf{e}_\theta \ ,$$

where $\mathbf{e}_r = -\cos\theta\,\mathbf{k} + \sin\theta\,\mathbf{i}$, $\mathbf{e}_\theta = \sin\theta\,\mathbf{k} + \cos\theta\,\mathbf{i}$.

ii. Suppose that the particle has mass m, and that the acceleration in part (a) arises from Newton's second law with a total force

$$\mathbf{F} \;=\; -mg\,\mathbf{k} + \mathbf{T} \ .$$

Show that

$$\mathbf{T}\cdot\mathbf{e}_r \;=\; -ml\dot{\theta}^2 - mg\cos\theta \ .$$

(b) i. Consider swinging on a swing with a chain of length l. Explain why the chain never becomes slack provided

$$-\cos\theta \;<\; \frac{l\dot{\theta}^2}{g}$$

holds throughout the motion, where θ is the angle the chain makes with the downward vertical.

ii. The swing initially hangs downwards, and a friend gives you a push in the horizontal direction with initial speed v. Using conservation of energy, show that provided $v > \sqrt{5gl}$ you'll swing all the way over the top without the chain ever becoming slack.